SkillSpring – Tracking Implementation

Project: Course Discovery Analytics POC

Website: https://luthfauu.github.io/course-discovery-analytics/

1. Project overview

SkillSpring is a simple course-discovery website created as a web analytics POC. The website has three courses: Data Analytics, Web Development, and Cybersecurity.

The main aim of this project is to understand how users interact with the website, track important actions using Google Tag Manager (GTM), and collect the data in Google Analytics 4 (GA4) for further analysis.

2. Website setup

I created a simple webpage and hosted it using GitHub Pages. The website contains course cards with Learn More and Enroll Now options, along with a demo enrollment form.

I added the GTM container code to the index.html file, including the required code in the head and body sections. I also added custom events to track actions such as viewing course details, clicking Enroll Now, and submitting the form.

3. Connecting GA4 and GTM

I first created a GA4 account and property named Course Discovery Analytics POC. Then, I created a web container in GTM and added its container code to the website code like I mentioned above in Website setup.

After that, I created a web data stream in GA4 and noted the Measurement ID. Instead of adding this ID directly to the website, I used it to connect GA4 with GTM.

I created a Google tag in GTM using the GA4 Measurement ID and set its trigger to All Pages. This allows GA4 to collect data whenever users visit the website.

4. Setting Up Events & Triggers

I created custom-event triggers in GTM for the events added to the website. Here are the three events which are added to the website code -:

- view_course
- enroll_click
- generate_lead

These events are attached while clicking on ‘Learn More’, ‘Enroll Now’ and ’Submit Enrollment’ buttons respectively. For each event, I created a GA4 event tag with the matching event name and connected it to its respective trigger.

For example, the GA4 - Event - view_course tag uses the CE_view_course trigger. I followed the same process for the other two events as well.

5. Testing and publishing

To identify which course a user interacts with, I created a Data Layer Variable in GTM.

The variable was configured as follows:

- Variable name: DLV_course_name
- Variable type: Data Layer Variable
- Data Layer Variable Name: course_name

I added course_name as an event parameter in the GA4 event tags. This allows the course name to be sent along with the events.

6. Testing and publishing

I used GTM Preview mode to test the tags and check whether the events were firing the tags correctly. I also checked whether the course_name variable was returning the correct course value.

After testing, I checked GA4 Realtime to confirm that the website visits and custom events were being received. I also used Debug View from Data Display Tab to verify that the event parameters were coming through correctly.

Once everything was working, I published the GTM container by creating a new version with a name and description.

I also marked generate_lead as a key event in GA4 because it represents a successful demo form submission.

7. GA4 Custom Dimension

To analyze the events based on the course name, I registered a custom dimension in GA4.

The configuration was:

- Dimension name: Course name
- Scope: Event
- Event parameter: course_name

This allows me to use the course name in GA4 Explorations and compare the events across different courses.

8. GA4 Reports and Explorations

I then created a funnel exploration named **SkillSpring Course Enrollment** to understand how users progress through the enrollment process. The funnel contains three steps and I named them as follows. Also, I added the events to each step accordingly.

1. Viewed Course Details – view_course
2. Clicked Enroll Now – enroll_click
3. Successfully Submitted Form – generate_lead

The funnel helps me understand how many users drop off at each step and how many continue to the next step, so I can identify where users are leaving the enrollment process.

Course Performance Analysis

I created a Free form exploration named Course Performance Analysis to compare the events across the three courses. For this, I imported the custom dimension which we already set which is Course name from Dimensions tab and also imported event count from Metrics Tab.

The report uses:

- Rows: Course name
- Columns: Event name
- Values: Event count

I applied a filter to include only the three main custom events: view_course, enroll_click, and generate_lead.

Device Analysis

I also created a Device Analysis report to explore how users interact with the website across different devices.

The report uses:

- Rows: Device category
- Values: Total users and Event count

9. Data Export and Analysis

I exported the filtered GA4 report as a CSV file for further analysis.

Since this is a small POC without a login system or a BigQuery event-level export, the actual GA4 data is mainly used to validate the tracking setup and explore the collected reports.

For deeper analysis using SQL and Python, I created a separate synthetic event-level dataset with simulated user IDs. This dataset is kept separate from the actual GA4 export and will be clearly labelled as synthetic data.

10. Limitations

This is a demo website, and the enrollment form is not connected to a real backend. The website also does not have a login system, so it does not use account-based user IDs.

The actual GA4 data is limited and may include test visits. Therefore, the findings from this data should not be treated as representative of real customer behavior.

The synthetic dataset will be used separately for further SQL and Python analysis, with its assumptions and limitations clearly mentioned.

11. Screenshots

Screenshots are kept in a separate Screenshots folder organised into GTM and GA4 subfolders.

Website link and corresponding html code is also placed in separate folders.

Suggested screenshots: GTM tags, trigger, variable, and Preview; GA4 Realtime, DebugView, custom dimension, funnel, Course Performance Analysis, and Device Analysis.